CSA4707-Deep Learning

Experiment 3: Multiclass Confusion Matrix

Aim:

To analyse the performance of a multiclass confusion matrix using a chosen dataset with Python.

Program:

```
from sklearn.datasets import load_digits from
sklearn.model_selection import train_test_split from
sklearn.ensemble import RandomForestClassifier from
sklearn.metrics import confusion_matrix, accuracy_score
import seaborn as sns import matplotlib.pyplot as plt

X, y = load_digits(return_X_y=True)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=23
)

clf = RandomForestClassifier(random_state=23)
clf.fit(X_train, y_train)

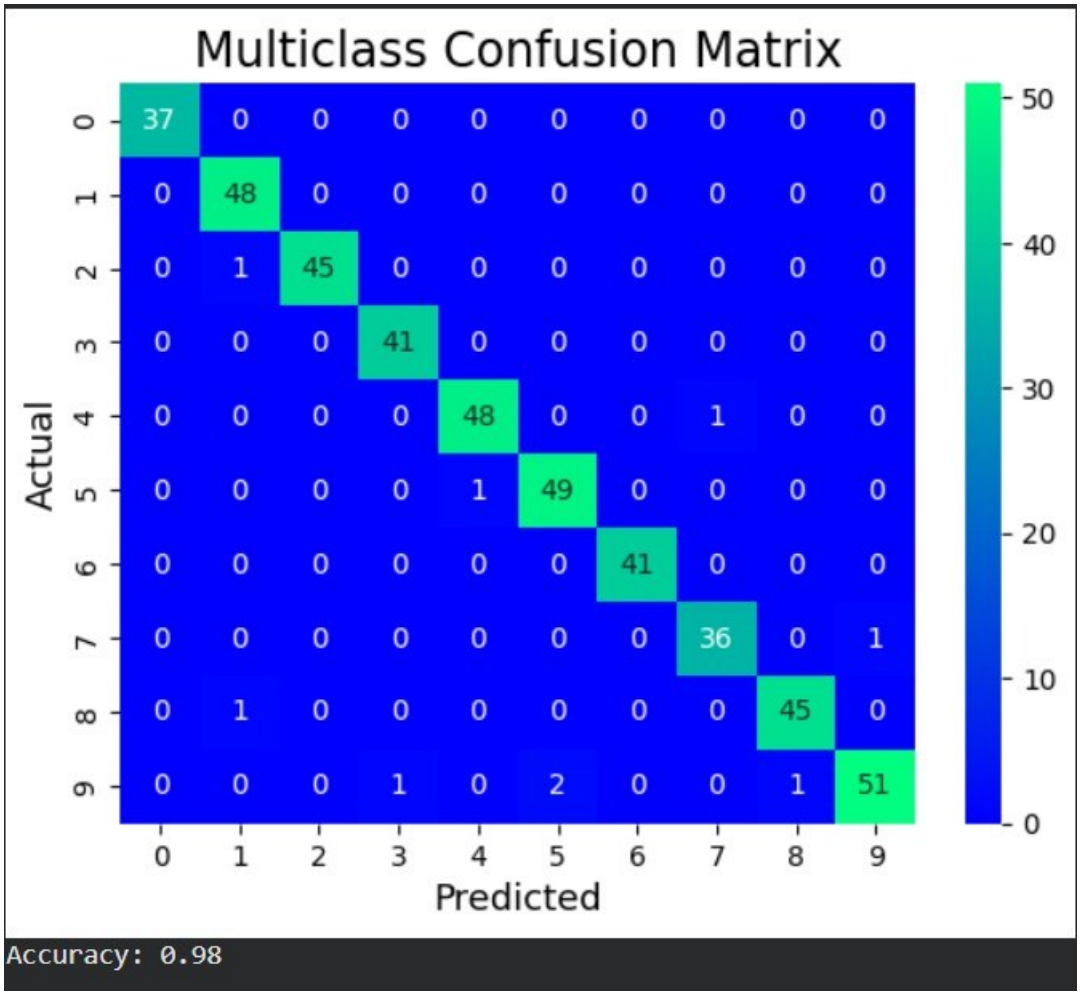
y_pred = clf.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='g', cmap="winter")
plt.ylabel('Actual', fontsize=13) plt.xlabel('Predicted',
fontsize=13) plt.title('Multiclass Confusion Matrix',
fontsize=17) plt.show()

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

Output:



Result / Output:

Accuracy: approximately 0.97 to 0.98

The multiclass confusion matrix was generated successfully for the handwritten digit dataset. The Random Forest classifier predicts the digit classes with high accuracy.